might be a way of characterizing our ignorance of what was really going on, rather than being the total truth about what is going on, as we're advocating here. One of the first things people did when learning about the uncertainty principle was to try to find loopholes in it. They failed, but in doing so we learned a lot about how quantum reality is fundamentally different from the classical world we had been used to.

The absence of definite quantities at the heart of reality that map more or less straightforwardly onto what we can eventually observe is one of the deep features of quantum mechanics that can be hard to accept upon first encounter. There are quantities that are not merely unknown but do not even exist, even though we can seemingly measure them.

Quantum mechanics forces us to confront this yawning chasm between what we see and what really is. In this chapter we'll see how that gap manifests itself in the uncertainty principle, and in the next chapter we'll see it again more forcefully in the phenomenon of entanglement.

The uncertainty principle owes its existence to the fact that the relationship between position and momentum (mass times velocity) is fundamentally different in quantum mechanics from what it was in classical mechanics. Classically, we can imagine measuring the momentum of a particle by tracking its position over time, and seeing how fast it moves. But if all we have access to is a single moment, position and momentum are completely independent from each other. If I tell you that a particle has a certain position at one instant, and I tell you nothing else, you have no idea what its speed is, and vice versa.

Physicists refer to the different numbers we use to specify something as that system's "degrees of freedom." In Newtonian mechanics, to tell me the complete state of a bunch of particles, you have to tell me the position and momentum of every one of them, so the degrees of freedom are the positions and the momenta. Acceleration is not a degree of freedom, since it can be calculated once we know the forces acting on the system. The essence of a degree of freedom is that it doesn't depend on anything else.

When we switch to quantum mechanics and start thinking about Schrödinger's wave functions, things become a little different. To make a